

- Carefully add two drops of dish soap to the container.
- Now, tilt the container 45 degrees and add 20 drops of chilled alcohol. Make sure the drop goes touching the inner wall of the container so as not to disturb the solution. Being less dense than the solution, alcohol will sit atop it.
- Put the lid on the container and gently turn it upside down and back to normal three times. Don't let bubbles form when you do this.
- Let the container sit undisturbed



You can preserve your DNA!

- for a minute or so. You'll see a milky thread is beginning to form, with tiny bubbles in it. That's your DNA!
- Insert the skewer and gently twist the DNA around it. You can preserve your DNA in a small vial with few drops of alcohol almost for eternity!

02 HOMEMADE SPECTROSCOPE

Light originating from different sources can be split into different wavelengths and their spectrums can be studied. Spectroscopes have long helped astrophysicists in determining the constituents of stars and far away galactic bodies, by analysing the light structure and spectrum details which allowed identifying the elements. Simply put, a spectroscope tells us what stars are made of! We'll tell how to make a miniature and simplified version of it. Gather these items:

- Empty paper towel roll (or any cardboard roll)
 - Scissors/knife
 - A CD/DVD
 - Pencil
 - Small cardboard piece
 - Tape
- Now it's time starting building our

home-made spectroscope:

- Take the knife and cut a thin slit of 45 degrees toward near the bottom of the roll
- Diametrically opposite the slit, cut a small rectangle that will act as your peephole/view hole.
- Trace one end of the roll on to the cardboard piece (basically make a circle) and cut it out.



Cut out a rectangular slit on the cardboard circle

- Cut out a rectangle slit on the cardboard circle you just made and tape it on the top of the roll, opposite end from where you made the cut.
 - Now insert the CD in the 45-degree slit, facing upwards.
- And, you are done! Point the spectroscope's upper slit towards the



This is what your finished home-made spectroscope should look like

sky and when you'll see through the view hole, you can feast your eyes on a rainbow. Go ahead and try with different light sources like neon lights, fluorescent lights, etc., and compare what you see!

03 HOMEMADE SMARTPHONE PROJECTOR

Who doesn't love an evening of their favourite movie playing on a big theatresque screen? However, buying projectors and setting them up is quite a costly affair. What if we told you that a good old magnifying glass can enhance your movie experience to quite some extent? 15 minutes and you'll be done with your homemade smartphone projector! Of course, it won't produce the same picture quality as the big machines but this is something you made yourself and that's the greater reward. Gather the following list of items:



- Rectangular box
- Cardboard
- Magnifying glass
- Scissors
- Glue
- Foil (to cover the back of the box to improve the projection quality, this item is optional)

Everything required is quite readily available in an average house, so let's begin constructing our projector:

- First, you need to remove the glass from the handle of the magnifying glass. If that's not possible, just make a slit while making the holder for the glass in the box.
- Centre the glass on the box and trace the outline.
- Now, using a scissor cut out a circle, as traced by the outline so that the glass can be fixed there.
- Put some glue where the glass and the cardboard box touch each other.